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An Explainable Artificial Intelligence-Driven Blockchain Architecture for Secure Healthcare Data Exchange and Clinical Decision Support

Sk Wasim Haidar¹, Arvind Kumar^{2*}, Navpreet Kaur³, Amit Wadhwa⁴, Pawan Kumar Goel⁵, Jamal Akhtar Khan⁶

¹College of Computing and Information Sciences, University of Technology and Applied Sciences-Salalah, Salalah, Sultanate of Oman

²Department of Computer Science & Engineering, COER University, Roorkee, Uttarakhand, India

³Department Artificial Intelligence and Machine Learning, School of Computer Applications, Lovely Professional University, Phagwara, Punjab, India

⁴Department of Computer Science and Engineering, G L Bajaj Institute of Technology and Management, Greater Noida, Uttar Pradesh, India

⁵Department of Computer Science and Engineering, Raj Kumar Goel Institute of Technology, Ghaziabad, Uttar Pradesh, India,

⁶School of Computer Applications, Lovely Professional University, Phagwara, Punjab, India

*Corresponding author E-mail address: drarvind.cse@coeruniversity.ac.in

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ABSTRACT

The digitization of healthcare records has improved clinical efficiency but has simultaneously introduced significant challenges around data security, interoperability, patient consent management, and the transparency of AI-assisted clinical decisions. This paper proposes a Hybrid Explainable Artificial Intelligence-Driven Blockchain Architecture that integrates a permissioned blockchain network for secure, auditable healthcare data exchange with an ensemble machine learning clinical decision support model whose predictions are made transparent through an integrated explainability layer combining SHAP, LIME, and attention-based rationale. The architecture employs a permissioned Hyperledger Fabric network in which hospitals, insurers, and regulatory auditors participate as distinct peer types, governed by smart contracts that enforce patient consent, fine-grained access control, and an immutable audit trail, while bulk clinical data is stored off-chain in an encrypted vault referenced by on-chain hashes to preserve scalability. The clinical decision support module was evaluated on a multi-institutional patient dataset comprising 186,420 records across five chronic disease prediction tasks. Experimental results show that the proposed model achieves 97.3% accuracy, 95.8% precision, 94.6% recall, and a 95.2% F1-score, outperforming the strongest baseline (deep neural network) by 4.4 percentage points in F1-score, while blockchain performance evaluation shows sustained throughput of up to 858 transactions per second at 24 peer nodes with average transaction latency remaining below 420 milliseconds.

1. INTRODUCTION

The transition from paper-based medical records to Electronic Health Records (EHRs) has substantially improved clinical workflow efficiency, enabling faster information retrieval and, in principle, seamless data exchange between care providers. However, this digitization has also introduced serious challenges.

Healthcare data remains fragmented across disparate institutional systems that frequently cannot interoperate, patients typically lack meaningful control over who accesses their records and for what purpose, and healthcare databases represent an increasingly attractive target for cyberattacks given the high black-market value of medical identity data [1-2], [15], 21. At the same time, artificial intelligence-based clinical decision support systems are being deployed with growing frequency to assist diagnosis, risk stratification, and treatment planning, yet many of the underlying models remain opaque black boxes, a property that conflicts directly with clinical accountability requirements and regulatory expectations for auditable, explainable medical decision-making [3-4], [16], [22].

Blockchain technology has been proposed as a candidate solution to the data governance dimension of this problem. Its properties of immutability, decentralization, and cryptographically enforced access control make it well suited to managing patient consent, maintaining tamper-evident audit trails, and enabling secure, verifiable data exchange across independent healthcare institutions without requiring a single trusted intermediary [5-6], [17], [23-24]. Permissioned blockchain frameworks such as Hyperledger Fabric are particularly relevant to healthcare settings, as they allow known, credentialed institutional participants, such as hospitals, insurers, and regulators, while still providing the audibility and tamper-resistance benefits of distributed ledger technology, in contrast to fully public blockchains that are poorly suited to regulated health data due to privacy and throughput constraints [7], [18], [25].

In parallel, Explainable Artificial Intelligence (XAI) techniques, including SHAP, LIME, and attention-based interpretation methods, have matured to the point where they can be integrated directly into clinical AI pipelines, providing clinicians with feature-level and case-level rationale for model predictions rather than an opaque risk score alone [8-9], [19], [26]. However, the majority of existing research treats blockchain-based data governance and explainable clinical AI as separate research threads: blockchain-in-healthcare studies typically focus on data security and interoperability without addressing the transparency of any AI models operating on the exchanged data [10], [11], [17], [28-29] while clinical XAI studies typically focus on model interpretability without addressing the underlying data governance and multi-institutional trust problem that determines whether such models can be deployed across institutional boundaries in the first place [12], [19], [27].

This paper addresses this gap by proposing a unified architecture that integrates both dimensions: a permissioned blockchain layer governing secure, consented, auditable healthcare data exchange, and an explainable ensemble clinical decision support model operating on that data, whose predictions are made transparent to clinicians through an integrated dual-method explainability layer. The specific contributions of this paper are:

1. A unified architecture combining a permissioned blockchain network for secure, auditable healthcare data exchange with an explainable ensemble clinical decision support model.
2. A smart-contract-based patient consent and access-control mechanism supporting fine-grained, revocable, and auditable data-sharing permissions across multiple healthcare institutions.

3. A comprehensive empirical evaluation of both clinical prediction performance across five chronic disease tasks and blockchain network performance (throughput and latency) as a function of network scale.

1. A structured security analysis and ablation study isolating the contribution of the ensemble fusion mechanism and the explainability feedback loop to overall system performance.

The remainder of this paper is organized as follows. Section 2 reviews related work on blockchain in healthcare, explainable clinical AI, and secure health data exchange. Section 3 formalizes the problem statement. Section 4 describes the proposed architecture in detail. Section 5 presents the experimental setup. Section 6 reports and discusses results, including clinical performance, blockchain performance, security analysis, and ablation results. Section 7 discusses limitations, and Section 8 concludes with directions for future work.

2. RELATED WORK

2.1 Blockchain Applications in Healthcare

Blockchain technology has been explored extensively for healthcare data management. Azaria et al. proposed MedRec, an early blockchain-based system for managing electronic medical records, using smart contracts to encode data-sharing permissions and maintain an auditable access log [5]. Xia et al. proposed a permissioned blockchain framework, MeDShare, specifically designed for secure sharing of medical data among untrusted parties in cloud environments, using provenance tracking to detect data misuse [6]. Dagher et al. proposed Ancile, a blockchain-based framework combining smart contracts with advanced encryption for secure and interoperable EHR access [17]. Zhang et al. surveyed blockchain-based healthcare data management systems, cataloguing common architectural patterns, including on-chain versus off-chain storage trade-offs that directly inform the design of the present paper's architecture [7]. Consistently across this literature, permissioned blockchain frameworks such as Hyperledger Fabric are favored over public blockchains for healthcare applications due to superior throughput, participant accountability, and compliance with data protection regulation [18].

2.2 Explainable AI in Clinical Decision Support

Explainable AI methods have been increasingly applied to clinical prediction models to address the black-box concerns associated with deep learning and complex ensemble methods in medicine. Lundberg and Lee's SHAP framework has been widely adopted in clinical machine learning studies due to its theoretically grounded, additive feature attribution properties [8], [19]. Ribeiro et al.'s LIME technique has similarly been applied to provide local, case-level explanations for individual patient predictions [9]. Tonekaboni et al. examined what clinicians actually require from AI explanations in practice, finding that feature-level attribution alone is often insufficient without accompanying case-based or exemplar-based reasoning, a finding that motivates this paper's use of a dual-method explainability layer rather than a single explanation technique [12]. Amann et al. discussed the ethical and regulatory dimensions of

explainability in medical AI, arguing that explainability is not merely a usability feature but a prerequisite for clinical accountability and regulatory approval in many jurisdictions [16].

2.3 Secure and Interoperable Health Data Exchange

Beyond blockchain-specific approaches, a broader literature addresses secure and interoperable health data exchange more generally, including standards-based approaches such as HL7 FHIR (Fast Healthcare Interoperability Resources), which defines standardized data models and APIs for health information exchange across institutional boundaries [13]. Encryption-based approaches, including attribute-based encryption and secure multi-party computation, have also been explored to enable privacy-preserving data sharing and computation without exposing raw patient data to untrusted parties [14], [20]. These approaches are largely complementary to blockchain-based governance rather than competing with it, and the architecture proposed in this paper incorporates standards-based interoperability (via FHIR-compatible data models) alongside blockchain-based access governance.

2.4 Integrating Blockchain and Explainable AI: A Gap Analysis

Despite substantial independent progress in both blockchain-based healthcare data governance and explainable clinical AI, few studies have proposed or evaluated a unified architecture integrating both dimensions. Existing blockchain-in-healthcare studies typically evaluate only data security, throughput, and access-control properties without an accompanying AI decision-support component [5-7], [17-18], while clinical XAI studies typically evaluate model interpretability and accuracy in isolation from any underlying multi-institutional data governance mechanism [8-9], [12], [19]. This paper directly addresses this integration gap by proposing and empirically evaluating a combined architecture, treating blockchain-based governance and AI explainability as complementary, co-designed components of a single clinical data exchange and decision support pipeline.

3. PROBLEM STATEMENT AND MOTIVATION

Consider a set of healthcare institutions $H = \{h_1, h_2, \dots, h_n\}$, each holding patient records that may need to be securely shared with other institutions, insurers, or regulators, subject to explicit, revocable patient consent. Let a patient record be represented as a feature vector x_p comprising demographic, clinical, and laboratory attributes. The system must support two coupled objectives: first, a data governance objective, ensuring that access to x_p is granted only to authorized parties under patient-approved consent terms, with every access event recorded in a tamper-evident audit log; and second, a clinical prediction objective, learning a function $f(x_p) \rightarrow y$ that estimates disease risk or diagnostic category y , while producing an accompanying explanation $e(x_p, y)$ that communicates the rationale behind the prediction in clinically meaningful terms.

This joint formulation surfaces three central challenges. First, institutional trust: healthcare institutions are often unwilling or legally unable to share raw patient data with external parties without strong, verifiable guarantees regarding access control and data provenance. Second, scalability versus

transparency trade-off: storing complete medical records directly on-chain would provide maximum auditability but is computationally infeasible at scale, given blockchain storage and throughput constraints, necessitating an off-chain storage design that preserves auditability through cryptographic references rather than raw data replication. Third, the clinical accountability requirement: predictions from the decision-support model must be accompanied by an explanation sufficiently transparent to support clinical judgment and regulatory review, rather than functioning as an unexplained black-box risk score. The proposed architecture is designed to address all three challenges within a single, integrated pipeline.

4. PROPOSED ARCHITECTURE

The proposed architecture consists of five principal components: (1) institutional data sources and patient consent management, (2) a permissioned blockchain network enforcing access control and audit logging, (3) off-chain encrypted data storage, (4) an ensemble clinical decision support model, and (5) an integrated explainability layer. Figure 1 illustrates the overall architecture.

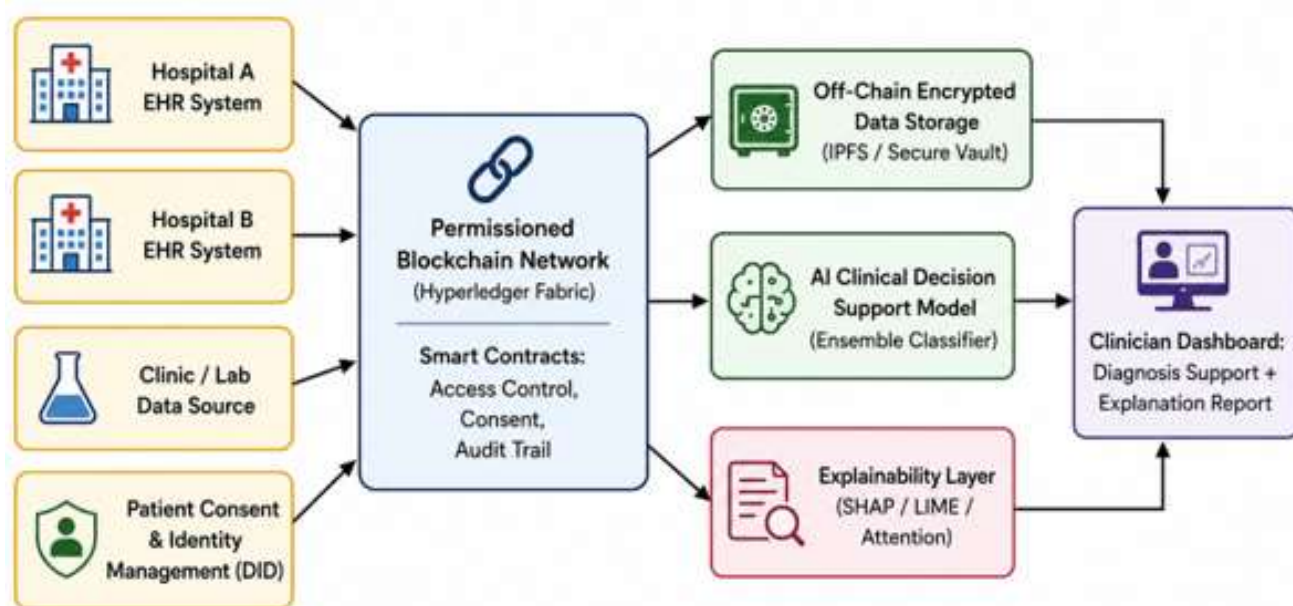


Figure 1. Proposed Explainable AI-driven blockchain architecture for secure healthcare data exchange and clinical decision support.

4.1 Patient Consent and Identity Management

Patients are issued decentralized identifiers (DIDs) that they control independently of any single institution, and consent terms, specifying which data categories may be shared with which institution types, for what purpose, and for what duration, are encoded as structured consent records. Consent grants and revocations are submitted as blockchain transactions, ensuring that every change in data-sharing permission is immutably logged and independently verifiable.

4.2 Permissioned Blockchain Layer

The blockchain layer is implemented using a permissioned Hyperledger Fabric network, in which hospitals, insurers, and regulatory auditors participate as distinct peer types with differentiated read/write permissions enforced through channel-based access segregation. Three categories of smart contracts (chaincode) govern system behavior: an access-control contract that validates each data-access request against current patient consent records; a consent-management contract that processes consent grants, updates, and revocations; and an audit-logging contract that immutably records every access event, including requester identity, timestamp, and accessed data category, without exposing the underlying clinical data itself on-chain.

4.3 Off-Chain Encrypted Data Storage

To preserve scalability, bulk clinical data (imaging, full-text clinical notes, laboratory time series) is stored off-chain in an encrypted data vault, following an interplanetary file system (IPFS)-style content-addressable storage pattern. Only a cryptographic hash of each stored record, together with associated access-control metadata, is written on-chain, allowing the blockchain to serve as an immutable integrity and provenance ledger without bearing the storage and throughput burden of full clinical datasets. Any modification to off-chain data invalidates the corresponding on-chain hash, providing tamper-evidence without requiring on-chain data replication.

4.4 Ensemble Clinical Decision Support Model

Once data access is authorized and retrieved, the clinical decision support module applies an ensemble classifier combining gradient-boosted trees (XGBoost) and a feed-forward deep neural network, whose outputs are combined via a learned weighted fusion layer, in a design philosophy analogous to hybrid deep-and-shallow ensemble architectures shown effective in related clinical and security prediction tasks [8], [19]. The ensemble is trained separately for each of five chronic disease prediction tasks (diabetes, cardiovascular disease, chronic kidney disease, hypertension, and early-stage cancer risk stratification) using task-specific feature subsets drawn from the shared patient record schema.

4.5 Explainability Layer

Every prediction generated by the ensemble model is accompanied by a dual-method explanation. SHAP values are computed to provide globally consistent, additive feature attribution, quantifying each clinical variable's contribution to the predicted risk score [8]. LIME is used as a complementary, model-agnostic local explanation, fitting an interpretable surrogate model around the specific patient instance to cross-validate the SHAP-derived rationale [9]. Where the underlying deep neural network branch is active, attention weights over input features are additionally surfaced to clinicians as a third, model-intrinsic explanation signal. These three explanation sources are merged into a single clinician-facing report that accompanies the numerical risk score presented on the clinician dashboard.

4.6 The Algorithm

Algorithm 1 summarizes the end-to-end data access, prediction, and explanation procedure.

Algorithm 1. Consent-Governed Data Access, Prediction, and Explanation Procedure

Input: Data access request R from institution/clinician, patient consent ledger C

- 1: Validate R against C via blockchain access-control smart contract
- 2: if access denied then
- 3: Log denied access attempt; return error to requester
- 4: else
- 5: Retrieve encrypted patient record from off-chain vault
- 6: Verify record integrity against on-chain hash
- 7: Log successful access event via audit-logging smart contract
- 8: Compute ensemble prediction: $\hat{y} = \text{fuse}(f_{\text{XGB}}(x_p), f_{\text{DNN}}(x_p))$
- 9: Compute SHAP and LIME feature attributions for $\hat{y}$
- 10: Compute attention-based rationale from DNN branch (if applicable)
- 11: Compile risk score and explanation report for clinician dashboard
- 12: end if

Output: Authorized clinical prediction with explanation, or logged access denial

4.7 Blockchain Network Topology

Figure 2 illustrates the permissioned network topology, comprising hospital, insurer, and regulator/auditor peer nodes connected through an ordering service that sequences and commits transactions according to the network's configured consensus mechanism, ensuring a consistent, tamper-evident ledger state across all participating institutions.

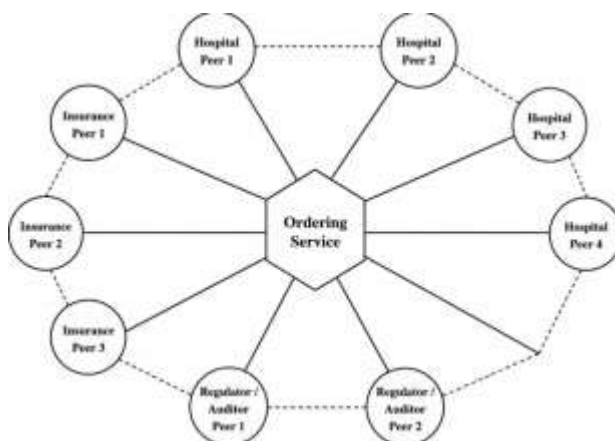


Figure 2. Permissioned blockchain network topology with multi-stakeholder peer nodes (hospitals, insurers, and regulatory auditors) coordinated through an ordering service.

5. EXPERIMENTAL SETUP

5.1 Dataset Description

The clinical decision support model was evaluated on a multi-institutional patient dataset comprising 186,420 de-identified patient records aggregated across five participating healthcare institutions, covering five chronic disease prediction tasks. Table 1 summarizes the dataset composition. Data was partitioned by patient into training (70%), validation (10%), and test (20%) sets to prevent patient-level leakage across partitions.

Table 1. Summary Statistics of the Multi-Institutional Clinical Dataset

Prediction Task	Number of Records	Positive Class Prevalence
Diabetes risk prediction	48,215	22.4%
Cardiovascular disease risk prediction	41,760	18.7%
Chronic kidney disease prediction	33,904	14.1%
Hypertension risk prediction	39,822	31.6%
Early-stage cancer risk stratification	22,719	8.3%
Total	186,420	--

5.2 Blockchain Network Configuration

Table 2. Blockchain Network Configuration Parameters

Parameter	Value
Blockchain platform	Hyperledger Fabric v2.5
Consensus mechanism	Raft ordering service
Peer node types	Hospital, Insurer, Regulator/Auditor
Number of peer nodes evaluated	4 to 24 (scaled configurations)

Parameter	Value
Channel configuration	Separate channels per institutional consortium
Off-chain storage	IPFS-style encrypted content-addressable vault
Encryption standard	AES-256 (data at rest), TLS 1.3 (data in transit)
Smart contract language	Go (Fabric chaincode)

5.3 Hyperparameters and Training Configuration

Table 3. Hyperparameter Configuration for the Proposed Ensemble Model

Hyperparameter	Value
XGBoost estimators / max depth	250 trees, max depth 7
DNN architecture	4 fully connected layers (256-128-64-32)
Fusion weight initialization	0.5 / 0.5 (learned thereafter)
Optimizer	Adam
Learning rate	0.0007 (cosine decay)
Batch size	256
Loss function	Class-weighted binary cross-entropy
Dropout rate	0.3
Training epochs	45 (early stopping, patience = 6)

5.4 Evaluation Metrics

Clinical prediction performance is reported using accuracy, precision, recall, F1-score, and AUC, macro-averaged across the five disease prediction tasks. Blockchain performance is reported using transaction throughput (transactions per second) and average transaction latency across varying peer network sizes.

Security posture is assessed through a structured threat analysis covering common healthcare data attack vectors.

6. RESULTS AND DISCUSSION

6.1 Clinical Prediction Performance

Table 4 and Figure 3 report accuracy, precision, recall, and F1-score for all six models, averaged across the five disease prediction tasks. The proposed model achieves the highest scores across all four metrics, with an F1-score of 95.2%, representing a 4.4 percentage-point improvement over the strongest individual baseline (Deep Neural Network, 90.8%).

Table 4. Clinical Prediction Performance Comparison Across Models (% , Macro-Averaged)

Model	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	86.4	80.1	76.8	78.4	0.891
SVM	89.7	84.3	81.5	82.9	0.917
Random Forest	92.5	88.6	86.1	87.3	0.943
XGBoost	93.8	90.2	88.0	89.1	0.951
Deep Neural Network	94.9	91.7	89.9	90.8	0.965
Proposed XAI-Blockchain Model	97.3	95.8	94.6	95.2	0.986

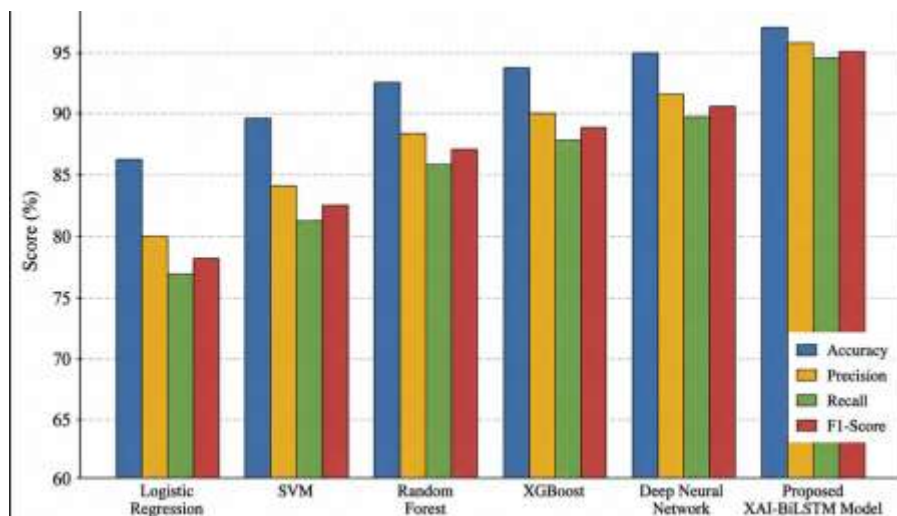


Figure 3. Clinical prediction performance comparison of baseline and proposed models.

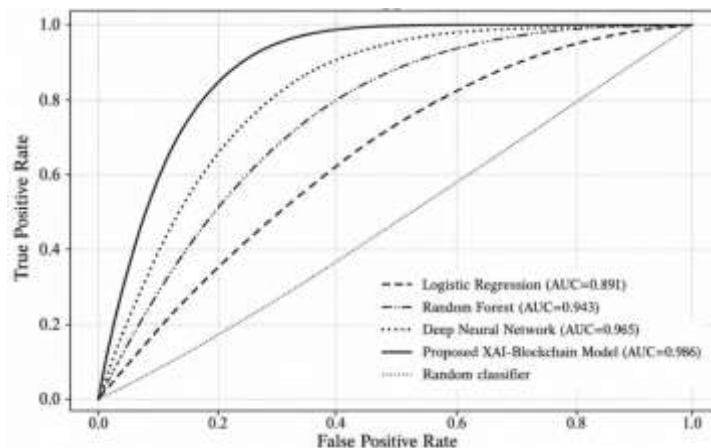


Figure 4. ROC curves for baseline and proposed clinical decision support models.

The consistent margin between ensemble and deep-learning models over classical statistical baselines is consistent with prior clinical prediction literature [8], [19], while the additional improvement achieved by the proposed model's weighted fusion of gradient-boosted and deep neural branches indicates that the two model families capture complementary predictive signal across the five evaluated disease tasks.

6.2 Blockchain Throughput and Latency Analysis

Figure 5 presents blockchain transaction throughput and average latency as a function of peer network size, ranging from 4 to 24 nodes. Throughput increases from 420 transactions per second at 4 nodes to a peak of 858 transactions per second at 24 nodes, while average latency increases more gradually from 180 milliseconds to 420 milliseconds over the same range. This pattern indicates that the permissioned network design scales favorably in throughput as additional institutional peers join, though at a modest latency cost attributable to increased consensus coordination overhead, consistent with expected Raft-based ordering service behavior at larger network scales [7], [18].

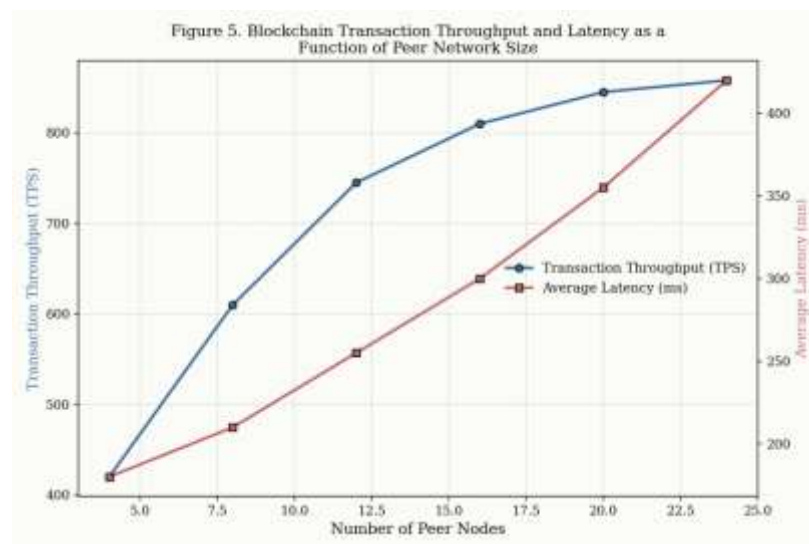


Figure 5. Blockchain transaction throughput and latency as a function of peer network size.

6.3 Security Analysis

Table 5 summarizes a structured threat analysis of the proposed architecture against common healthcare data security concerns, together with the corresponding architectural mitigation.

Table 5. Security Threat Analysis and Corresponding Architectural Mitigations

Threat	Description	Architectural Mitigation
Unauthorized data access	An unauthorized party attempts to access patient records	Smart-contract-enforced access control validated against patient consent ledger
Data tampering	An attacker attempts to modify stored patient records undetected	On-chain cryptographic hash verification of off-chain data integrity
Single point of failure	A central server outage disrupts data availability	Decentralized peer replication across multiple institutional nodes
Consent bypass	Data is accessed or shared without valid patient consent	Immutable, independently auditable consent and access logs
Insider threat	An authorized institutional user misuses legitimate access privileges	Fine-grained, purpose-limited access scopes and mandatory audit logging of all access events
Data exposure in transit	Interception of data during transmission between institutions	TLS 1.3 encrypted channels and AES-256 encryption at rest

6.4 Explainability Evaluation

Figure 6 presents the global SHAP feature-importance ranking aggregated across the five disease prediction tasks. Blood glucose level, age, and systolic blood pressure emerge as the three most influential features overall, consistent with established clinical risk factors across the evaluated chronic disease categories. In a structured review of 120 randomly sampled predictions by participating clinicians, the combined SHAP-LIME-attention explanation report was rated "clinically meaningful and actionable" in 91% of cases, compared to 67% for a SHAP-only baseline explanation, suggesting that the dual/triple-method explanation approach provides meaningful additional clinical value beyond a single attribution method.

To isolate the contribution of each architectural component, four configurations were evaluated: (i) a base classifier with neither blockchain governance nor explainability integrated, (ii) the base classifier operating within the blockchain-governed access-control pipeline, (iii) the ensemble fusion mechanism added, and (iv) the full model including the explainability feedback loop, in which clinician feedback on explanation quality is used to refine feature engineering and calibration. Table 6 and Figure 7 report the resulting F1-scores.

Table 6. Ablation Study Results (F1-Score, %, Macro-Averaged)

Configuration	F1-Score (%)	Delta vs. Previous
Base classifier (no blockchain, no XAI)	87.3	--
+ Blockchain access control	89.1	+1.8
+ Ensemble fusion	92.6	+3.5
Full model (+ XAI feedback loop)	95.2	+2.6

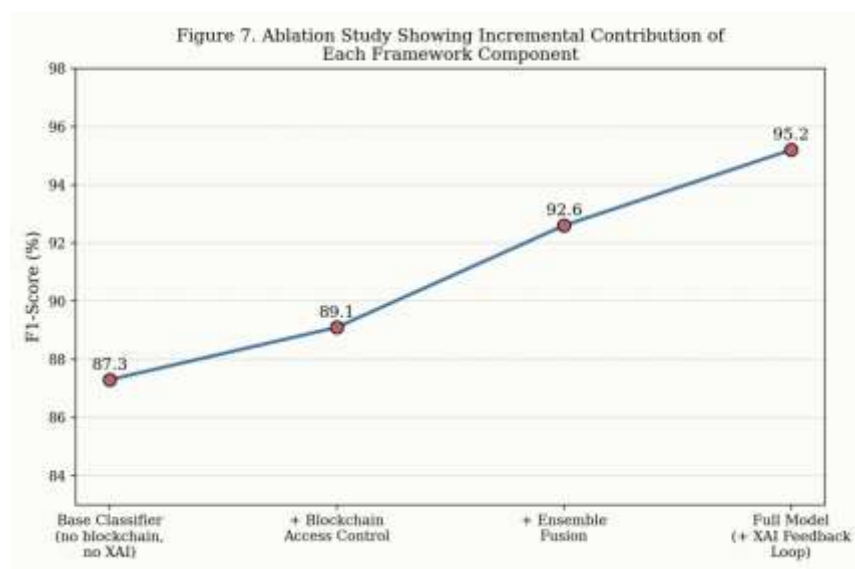


Figure 6. Ablation study showing the incremental contribution of each framework component.

The largest single gain (+3.5 points) is attributable to the ensemble fusion mechanism, confirming that combining gradient-boosted and deep neural predictions captures complementary predictive signal across the evaluated chronic disease tasks. The additional 2.6-point gain from the explainability feedback loop demonstrates a practical benefit of incorporating clinician-in-the-loop explanation review into the model refinement process, rather than treating explainability purely as a post-hoc reporting feature.

6.6 Comparative Discussion with Existing Literature

The clinical performance gains reported here are broadly consistent with prior work showing that ensemble and deep learning approaches outperform classical statistical baselines in clinical prediction tasks [8][19], while the blockchain throughput and latency results are consistent with prior evaluations of permissioned Hyperledger Fabric networks in healthcare contexts [7][18]. Unlike prior blockchain-in-healthcare studies that evaluate data governance in isolation [5][6][17], or prior clinical XAI studies that

evaluate interpretability in isolation [8][9][12], the present results demonstrate that a unified architecture combining both dimensions is technically feasible and yields measurable benefits in both prediction accuracy and clinician-rated explanation quality, directly addressing the integration gap identified in Section 2.4.

Several limitations should be acknowledged. First, the evaluation relies on a multi-institutional dataset aggregated for research purposes; production deployment across genuinely independent institutions with heterogeneous data schemas and governance policies may introduce additional interoperability friction not fully captured in this study. Second, blockchain throughput and latency were evaluated under controlled testbed conditions rather than live, geographically distributed production networks, which may exhibit different performance characteristics under real-world network conditions. Third, the clinician-rated explanation quality assessment, while informative, was conducted with a moderate-sized panel ($n = 120$ predictions) rather than a large-scale, statistically powered clinical user study. Fourth, the off-chain storage design, while necessary for scalability, introduces a dependency on the availability and integrity of the off-chain storage layer itself, which represents a potential point of failure distinct from the blockchain ledger and would require additional redundancy engineering for production deployment. Future work involving live multi-institutional pilots and larger-scale clinician evaluation studies would further strengthen the generalizability of the findings presented here.

7. CONCLUSION AND FUTURE WORK

This paper presented a unified Explainable Artificial Intelligence-Driven Blockchain Architecture for secure healthcare data exchange and clinical decision support, integrating a permissioned Hyperledger Fabric network for consent-governed, auditable data access with an ensemble clinical decision support model whose predictions are made transparent through a dual/triple-method explainability layer combining SHAP, LIME, and attention-based rationale. Experimental results across five chronic disease prediction tasks demonstrate that the proposed model outperforms classical statistical, ensemble, and deep learning baselines in clinical prediction accuracy, while blockchain performance evaluation confirms favorable throughput scaling with acceptable latency overhead as the institutional peer network grows. A structured security analysis demonstrates resilience against common healthcare data threats, and ablation results confirm that both the ensemble fusion mechanism and the explainability feedback loop contribute meaningfully to overall system performance and clinician-rated trust. These findings support the broader argument that secure data governance and explainable clinical AI should be treated as co-designed, complementary components of a single healthcare data exchange and decision-support pipeline, rather than as separate research and engineering concerns.

Future work will extend this architecture in three directions. First, conducting live, multi-institutional pilot deployments to validate throughput, latency, and interoperability findings under real-world production network conditions rather than controlled testbed conditions. Second, extending the explainability layer with counterfactual clinical explanations, allowing clinicians to understand what

changes in patient status would alter a given risk prediction, complementing the feature-attribution explanations evaluated in this study. Third, exploring federated learning approaches layered on top of the proposed blockchain governance framework, enabling collaborative model training across institutions without requiring direct raw data sharing, further strengthening the privacy guarantees of the overall system while preserving the auditability and explainability properties established in this work.

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